Practice Effects Confound the Emotion–Performance Link in Self-Regulated Learning: A Reanalysis of Lin et al. (2026)

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Abstract

Lin et al. (*Journal of Educational Psychology*, 2026) proposed a computational model of the dynamic interplay between goal setting, performance, and emotions in self-regulated learning. In a repetitive math task (Study 1), they reported that positive emotions predicted lower subsequent performance, interpreting this via the coasting hypothesis. We argue that this finding may instead reflect a temporal confound: in a repetitive task, performance improves due to practice while emotional engagement declines, producing a spurious negative association when time trends are unmodeled. We reanalyze the Study 1 data by introducing a log(trial) learning curve into the best-fitting model. The negative emotion-to-performance effect is no longer credible in this extended model; instead, a strong practice effect emerges. This confound-based interpretation also parsimoniously accounts for why the emotion-to-performance pathway reversed direction in the original Study 2, where task variety and high stakes attenuate practice-driven time trends. Methodologically, we show that the WAIC criterion used in the original study exhibits poor diagnostic reliability and is theoretically biased for autoregressive panel models. We implement exact leave-future-out cross-validation as an unbiased alternative, under which the original model selection conclusions are no longer supported.

Keywords: coasting hypothesis, computational modeling, leave-future-out cross-validation, model comparison, practice effects, self-regulated learning

Practice Effects Confound the Emotion–Performance Link in Self-Regulated Learning: A Reanalysis of Lin et al. (2026)

Self-regulated learning (SRL) describes the process by which learners actively monitor and adjust their goals, strategies, and effort in pursuit of academic outcomes (Panadero, 2017; Pintrich, 2000; Zimmerman, 2000). A central tenet of SRL frameworks is that learners set goals, evaluate their performance against those goals, and revise their behavior accordingly. Within this cycle, emotions have been increasingly recognized as playing a key role: how students feel about their progress influences how they set subsequent goals and how effectively they perform (Ben-Eliyahu, 2019; Efklides, 2011; Efklides & Schwartz, 2024; Pekrun, 2006).

A natural expectation, grounded in several theoretical traditions, is that positive emotions should facilitate cognitive performance. Control-value theory posits that positive activating emotions such as enjoyment enhance attention, motivation, and the use of flexible learning strategies, thereby supporting achievement (Forsblom et al., 2022; Pekrun et al., 2002; Pekrun, 2006). Similarly, the broaden-and-build theory suggests that positive affect broadens the scope of cognition and builds intellectual resources (Fredrickson, 2001). Empirical evidence from both laboratory and educational settings has largely supported this positive link (e.g., Berweger et al., 2025; Forsblom et al., 2022; Pekrun et al., 2017). However, some studies have reported null or context-dependent effects. For example, in a high-stakes examination context, high positive affect did not reliably predict better achievement once controls were introduced (Jerrim, 2022). Moreover, certain positive emotions may reduce subsequent effortful engagement: in daily academic goal striving, pride was associated with lower next-day effort (Seo & Patall, 2021). Experimental evidence also suggests that positive mood can impair working-memory performance under certain conditions (Martin & Kerns, 2011).

Recently, Lin et al. (2026) proposed a computational model of the dynamic interplay between goal setting, performance, and emotions in SRL. They applied their model to two empirical datasets: a controlled online math task in which participants solved three-digit arithmetic problems across 20 two-minute blocks (Study 1), and a naturalistic learning context in

which German medical students studied for a high-stakes state exam over a period of up to 40 days (Study 2). In Study 2, the model confirmed the expected positive association: stronger positive emotions predicted higher subsequent performance. In Study 1, however, the model estimated a negative emotion-to-performance link, suggesting that positive emotions led to worse subsequent performance. The authors interpreted this counterintuitive finding through the lens of the coasting hypothesis (Carver, 2003; Thürmer et al., 2020), which proposes that positive emotions signal that sufficient progress has been made toward one's goals, leading individuals to reduce effort and consequently perform worse. While this interpretation is theoretically coherent, the reversal of the effect across the two studies was not given a unified explanation.

However, in this paper, we propose that the negative emotion-to-performance association observed in Study 1 may reflect a temporal confound rather than a genuine causal process. There are at least two reasons to suspect that unmodeled time trends could generate such a confound in Lin et al.'s (2026) work. First, practice effects are well documented in repetitive cognitive tasks: repeated exposure to the same type of problem can increase fluency and improve performance over time (Newell & Rosenbloom, 2013; Scharfen et al., 2018; Tenison et al., 2016). Second, affective responses are also time-sensitive. Repeated stimulation can attenuate responsiveness through habituation (Rankin et al., 2009; Thompson & Spencer, 1966), while sustained engagement in a monotonous task may alter emotional experience through boredom and fatigue (e.g., Tanaka & Murayama, 2014). Recent intensive longitudinal evidence further suggests that emotional states differ in their temporal dynamics rather than following a single uniform pattern (Berweger et al., 2025). As the task progresses, multiple temporally driven processes may therefore operate simultaneously: performance may improve due to practice, while affective states may change over time because of habituation, boredom, and fatigue. If these trends are not accounted for in the model, the resulting covariance between rising performance and shifting emotions may be misattributed to a direct causal pathway from emotions to performance, producing a spurious negative coefficient.

A second concern relates to the model comparison procedure used to evaluate competing

specifications. A key finding of the original study was that Alternative 1, a model that distinguishes between the effects of goal success and goal failure on subsequent goal revision, outperformed the Hypothesized model (which treats the goal-performance discrepancy symmetrically) on the basis of WAIC. This comparison led the authors to conclude that piecewise modeling of the discrepancy is warranted. However, the article reported only raw WAIC values without standard errors or diagnostic indicators, making it impossible for readers to assess whether the observed difference was statistically meaningful. In the present study, we supplement the original comparison with two additional evaluation criteria. The first is PSIS-LOO cross-validation[1] (Vehtari et al., 2017), which improves on WAIC in several respects: (i) it is asymptotically equivalent to WAIC under regular conditions, while often being more robust in finite samples; (ii) it provides Pareto k diagnostics to flag unreliable estimates (Vehtari et al., 2015); and (iii) it supports stacking weights for quantifying relative model contributions (Yao et al., 2018). However, both WAIC and PSIS-LOO are biased for models with autoregressive structure, for two related reasons: (i) leaving out a single observation still allows future time points to inform predictions of the past, leaking temporal information into the estimate; and (ii) this leakage can systematically favor more highly parameterized models, because their additional flexibility gives them more opportunity to absorb temporally leaked information and thereby appear superior in model comparison. Our second criterion, exact leave-future-out cross-validation (LFO-CV, Bürkner et al., 2020), addresses this by using only past observations to predict future ones, thereby aligning model evaluation with the forecasting task of interest and avoiding the optimistic bias induced by temporal leakage.

In the following, we will then (1) replicate the original model fits for the Hypothesized and Alt1 models and extend Alt1 by introducing population-level log(trial) effects on goals, performance, and emotions, testing whether the negative emotion-to-performance effect is robust to the inclusion of a learning curve;[2] (2) evaluate model comparison across three criteria of

[1] Pareto-smoothed importance sampling leave-one-out cross validation, PSIS-LOO-CV.

[2] All data and analysis scripts are publicly available at https://github.com/lukezed/emoDAG_demo. We optimized the Stan implementation of the original models but the mathematical specifications are equivalent; see Appendix A.

increasing rigor (WAIC, PSIS-LOO, and exact LFO-CV); and (3) offer an alternative interpretation of the contradictory emotion-to-performance findings across Studies 1 and 2, grounded in the temporal confound identified above.

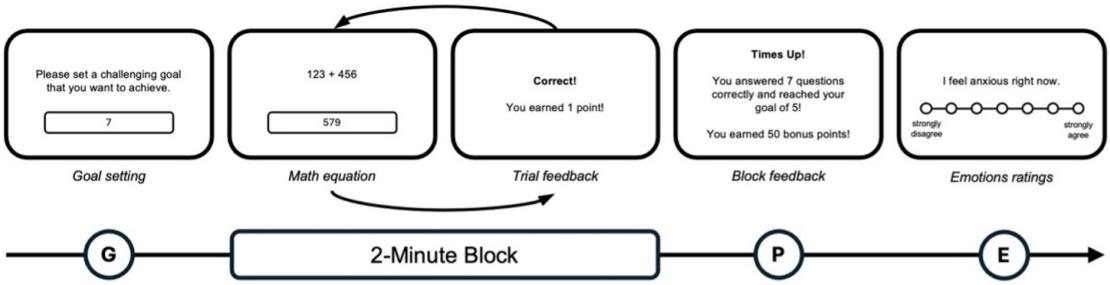
Method

Data and Task

We reanalyzed the publicly available Study 1 data from Lin et al. (2026). A total of 357 participants ($n_{\text{female}} = 212$, $M_{\text{age}} = 38.83$, $SD = 12.40$) completed an online math task consisting of 20 two-minute blocks of three-digit arithmetic problems within a single session (see Figure 1). At the start of each block, participants set a numerical goal; after each block, they received performance feedback and rated their emotional state. The openly available dataset contains three variables per participant per block: goal level (number of questions intended), performance (number of correct answers), and an aggregated emotion index ranging from approximately -3 to 3 in increments of $1/3$, where positive values indicate positive affect. The discrete emotion ratings from which this index was derived are not available in the open data. Following Lin et al. (2026), we treated this aggregated index as approximately continuous. For full details of the task design, incentive structure, and emotion aggregation procedure, see Lin et al. (2026).

Figure 1

Study 1 Experimental Design. Note. *Reprinted from "The Dynamic Interplay Between Goal Setting, Performance, and Emotions in Self-Regulated Learning: A Computational Modeling Approach," by W. M. Lin, L. FitzGibbon, M. Theobald, J. Breitwieser, G. Brod, K. Murayama, and M. Sakaki, 2026,* Journal of Educational Psychology. *Participants were prompted to set a goal for the number of questions they wanted to answer correctly (G), answered as many math questions correctly as possible within a two-minute block, after which their performance was shown (P), and then rated their emotions on a 7-point Likert scale (E). This sequence was repeated for 20 blocks.*



Computational Models

All three models share the same general structure: at each time point, goals (G), performance (P), and emotions (E) are modeled as jointly normally distributed, conditional on values at the previous time point and on cross-lagged predictors. Block 1 served as the initial condition and was not itself modeled. Full mathematical specifications are provided in Appendix A; here we describe the substantive interpretation of each model.

Hypothesized Model

This model follows the original specification of Lin et al. (2026) and embodies a symmetric self-regulation process. At each time step, goal revision is driven by three forces: (i) an autoregressive tendency (goals persist over time), (ii) the goal-performance discrepancy (GPD; the difference between what the participant achieved and what they aimed for), and (iii) the participant's emotional state. Crucially, the GPD enters the goal equation with a single coefficient, meaning that exceeding one's goal by two questions has the same magnitude of effect on subsequent goal revision as falling short by two questions. Performance at each time step is predicted by the previous performance, the discrepancy between the newly set goal and the previous performance, and the previous emotional state. Emotions are predicted by their own autoregressive component and by the GPD computed after the current performance is observed. This model has 11 individual-level parameters per participant.

Alt1 Model

Alt1 modifies only the goal equation. The single symmetric GPD coefficient is replaced by two separate coefficients: one governing the effect of goal success (when performance exceeds the goal) and one governing the effect of goal failure (when performance falls short). All other equations remain identical to the Hypothesized model. This was the best-fitting model reported by Lin et al. (2026), and its superiority over the Hypothesized model on the basis of WAIC was taken as evidence that success and failure exert asymmetric effects on goal revision. This model has 12 individual-level parameters per participant.

LogTrial Model

This model extends Alt1 by adding three population-level log(trial) covariates, one to each of the goal, performance, and emotion equations. These terms allow each outcome to absorb a parsimonious nonlinear time trend over the course of the 20 blocks, such as practice-related improvement in performance, gradual goal adjustment, or broad affective shifts over time. Aside from these three additional fixed effects, the model is identical to Alt1. The log-trial coefficients were specified at the population level only, without individual-level variation, as a parsimonious first-order correction for broad temporal trends. This choice also avoided the convergence problems we encountered when allowing these trends to vary by participant. This model has 12 individual-level parameters and 3 additional population-level fixed effects.

Estimation

Parameters were estimated in a hierarchical framework using weakly informative $\mathcal{N}(0, 3)$ priors. For the Hypothesized and Alt1 models, prior specifications and parameter structure were matched to Lin et al. (2026). Our Stan implementation differed only in computational details, most notably the use of a non-centered parameterization to improve sampling efficiency; the underlying mathematical model is equivalent, and the recovered parameter estimates closely reproduced the original results.

All models were fit in `Stan` via the `CmdStanR` interface (Gabry et al., 2025), with 4 chains of 1,000 warmup and 1,000 sampling iterations each (4,000 posterior draws total), `adapt_delta = 0.95`, and `max_treedepth = 15`. All models converged without divergences, no transitions hit the maximum treedepth, all $\hat{R}$ values were below 1.01, and bulk and tail effective sample sizes were adequate for all monitored parameters.[3]

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[3] The original study specified 10,000 iterations per chain with `adapt_delta = 0.99` and `max_treedepth = 20`. These settings are considerably more conservative than necessary for the present implementation; our reduced settings were sufficient to produce stable posterior estimates that closely recovered the original parameter values for the Hypothesized and Alt1 models (see Results).

Model Comparison

As outlined in the Introduction, we evaluated model fit using three criteria of increasing suitability for autoregressive structures.

WAIC. We computed the widely applicable information criterion (Watanabe, 2010) from the pointwise log-likelihood using the `loo` R package (Vehtari et al., 2017). Following Vehtari et al. (2017), we examined the proportion of observations with estimated effective numbers of parameters ($\hat{p}_{\text{WAIC}}$) exceeding 0.4, which signals that the WAIC estimate for that observation may be unreliable.

PSIS-LOO. We computed Pareto-smoothed importance sampling leave-one-out cross-validation (Vehtari et al., 2015; Vehtari et al., 2017) from the same pointwise log-likelihood draws. Pareto k diagnostics were used to assess the reliability of each pointwise estimate: observations with $k > 0.7$ indicate that the importance sampling approximation may be unreliable. When only a small number of observations are flagged, remedies are available, including moment matching (Paananen et al., 2021) and refitting the model with the problematic observations excluded. However, when the number of problematic observations is large, Vehtari et al. (2017) recommend more robust alternatives. As reported in the Results, all three models exhibited a large number of problematic Pareto k values. This makes standard PSIS-LOO unreliable. Although K-fold cross-validation is often recommended in such cases, it would not solve the temporal leakage problem in the present autoregressive setting. We therefore used exact LFO-CV as our primary time-respecting evaluation criterion. We also computed stacking weights (Yao et al., 2018) to quantify each model's relative predictive contribution under PSIS-LOO.

LFO-CV. We implemented exact one-step-ahead leave-future-out cross-validation (Bürkner et al., 2020), adapted for our panel data structure. Starting from an initial history of $L = 5$ blocks (25% of the 20 blocks), we iteratively fit each model using data up to block t, with block 1 treated as the initial condition rather than a modeled observation, and evaluated one-step-ahead predictive performance for block $t + 1$ for all 357 participants simultaneously, yielding 15 one-step-ahead predictive evaluations (blocks 6 through 20). At each step, the model

was refit from scratch using the same MCMC settings as in the full-sample analysis. For each evaluation, we computed the per-block expected log predictive density (ELPD) by averaging the exponentiated log-likelihood across posterior draws for each participant and then summing the log of these averages across all $N = 357$ participants. The total LFO-CV score is the sum of per-block ELPDs across the 15 prediction steps. For pairwise model comparison, we computed ΔELPD as the difference in total LFO-CV scores. We also report an approximate standard error based on the variability of the 15 blockwise predictive differences, interpreted cautiously given the small number of forecasting steps. Full implementation details and code are available in the online reproducibility materials.

Results

Replication and Predictive Diagnostic Concerns

We first verified that our reimplementation reproduced the original WAIC-based comparison closely. For the Hypothesized and Alt1 models, our WAIC values were within the expected range of those reported by Lin et al. (2026) (Table 1, A). As in the original paper, Alt1 obtained a lower WAIC than the Hypothesized model, and adding log(trial) effects further improved WAIC for the LogTrial model.

However, these WAIC comparisons were not reliable. All three models showed a substantial proportion of observations with $\hat{p}_{\text{WAIC}} > 0.4$ (16.7%–17.4%; Table 1, A), indicating unstable pointwise WAIC contributions for a nontrivial share of the data. This alone weakens the original WAIC-based model-selection argument, especially because Lin et al. did not report further predictive diagnostics. We therefore extended the original analysis by examining PSIS-LOO as an additional check. As shown in Figure 2 and Table 1 (B), all three models produced numerous large Pareto k values, with 89–113 observations exceeding 0.7 and 19–25 exceeding 1.0. These results show that the standard approximate criteria were diagnostically problematic in this setting. For this reason, and because the data are temporally ordered, we treat exact LFO-CV as the primary basis for model comparison.

Figure 2

Pareto k *Diagnostic Values From PSIS-LOO Cross-Validation for the Hypothesized (a), Alt1 (b), and LogTrial (c) Models. Each point represents one of 6,783 observations. Dashed and solid horizontal lines mark the* k *= 0.7 and* k *= 1.0 thresholds, respectively. Yellow points (*k *> 0.7) indicate potentially unreliable importance sampling estimates; red points (*k *> 1.0) indicate that a finite mean of the importance weights does not exist. All three models exhibit problematic* k *values well in excess of what can be remedied through moment matching or pointwise refitting.*

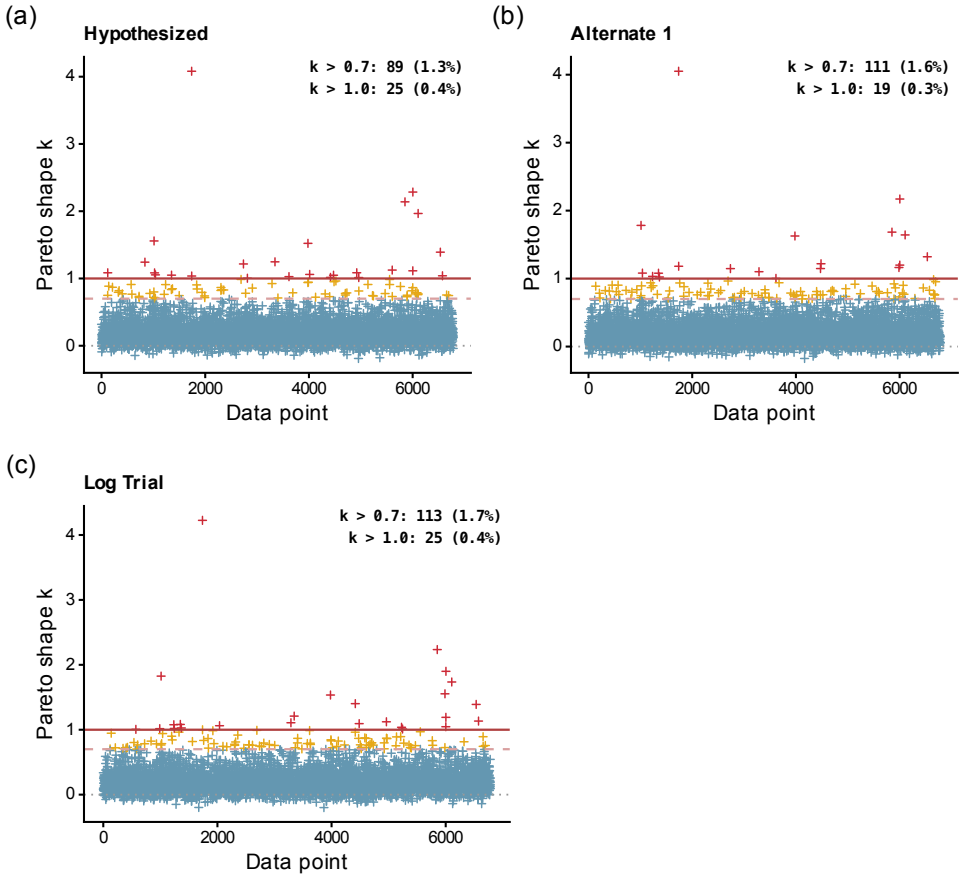


Table 1

Model comparison results for the three candidate models. In A, lower WAIC values indicate better fit. In B, $\Delta\text{ELPD}_{\text{LOO}}$ values are reported relative to the best-performing model, LogT, which is set to 0; higher Pareto k counts indicate less stable PSIS-LOO approximations; stacking weights are from the three-model comparison. In C, pairwise differences are reported so that positive values favor the first-named model; pairwise stacking weights are from the corresponding two-model comparisons and are reported for the first-named model.

	A. WAIC comparison		
Model	WAIC in Lin et al. (2026)	WAIC (SE)	$p_{\text{WAIC}} > 0.4$ (%)
Hyp	52,934.7	53,484.4 (1,149.8)	16.7%
Alt1	52,743.3	53,326.1 (1,176.6)	17.2%
LogT	—	53,095.0 (1,156.6)	17.4%

	B. PSIS-LOO diagnostics			
Model	$\Delta\text{ELPD}_{\text{LOO}}$ (SE)	$k > 0.7$ n (%)	$k > 1.0$ n (%)	Stacking weight (%)
Hyp	-177.2 (33.8)	89 (1.3%)	25 (0.4%)	17.7%
Alt1	-99.8 (12.4)	111 (1.6%)	19 (0.3%)	0.0%
LogT	0.0 (0.0)	113 (1.7%)	25 (0.4%)	82.3%

	C. Pairwise comparisons across different criteria			
Comparison	$\Delta\text{ELPD}_{\text{WAIC}}$ (SE)	$\Delta\text{ELPD}_{\text{LOO}}$ (SE)	$\Delta\text{ELPD}_{\text{LFO}}$ (SE)	First-model w_{stack} (%)
LogT − Hyp	194.7 (33.0)	177.2 (33.8)	78.12 (28.92)	85.8%
LogT − Alt1	115.5 (15.4)	99.8 (12.4)	39.54 (19.20)	100.0%
Alt1 − Hyp	79.7 (34.0)	77.5 (33.6)	38.57 (33.80)	70.1%

Model Comparison Across Criteria

Table 1 (C) presents pairwise comparisons under WAIC, PSIS-LOO, and exact LFO-CV. The clearest result is that the LogTrial model outperformed both the Hypothesized and Alt1 models under all three criteria. Under exact LFO-CV, LogTrial remained clearly better than the Hypothesized model ($\Delta\text{ELPD} = 78.12$, SE = 28.92). Relative to Alt1, its advantage was smaller but still significant ($\Delta\text{ELPD} = 39.54$, SE = 19.20).

The more consequential comparison is Alt1 versus the Hypothesized model, because this was the contrast on which the original model-selection conclusion depended. Under WAIC and PSIS-LOO, Alt1 outperformed the Hypothesized model ($\Delta\text{ELPD} = 79.7$ and 77.5, respectively). Under exact LFO-CV, however, this difference was reduced to $\Delta\text{ELPD} = 38.57$ with SE = 33.80. Alt1 therefore remained directionally preferred, but its predictive advantage was small relative to its uncertainty once the comparison was made with a temporally appropriate criterion. The

original conclusion that Alt1 should be selected over the Hypothesized model was thus not strongly supported.

The stacking weights tell the same story. In the three-model PSIS-LOO comparison, Alt1 received zero weight, whereas LogTrial received 82.3% and the Hypothesized model 17.7% (Table 1, B). In the pairwise comparison with the Hypothesized model, Alt1 received 70.1% of the stacking weight (Table 1, C). Thus, Alt1 is more likely than the Hypothesized model to be the better predictive specification, but the advantage is limited, and it disappears as a serious competitor once LogTrial model is included.

LFO Per-Block Comparisons

Figure 3 shows the per-block ΔELPD values from exact LFO-CV. The LogTrial model outperformed the Hypothesized model in 13 of the 15 prediction steps. Relative to Alt1, LogTrial performed better in 12 blocks, tied in 1, and performed worse in 2, yielding a smaller but still positive aggregate advantage.

The Alt1 versus Hypothesized comparison was weaker. Alt1 performed better in 11 blocks, tied in 1, and performed worse in 3, but much of its cumulative advantage was offset by the Hypothesized model's especially strong performance in block 10. The blockwise pattern is therefore consistent with the aggregate LFO-CV result: Alt1 tended to perform better, but not by a margin that was clearly separated from its uncertainty.

Parameter Estimates

Table 2 presents the population-level parameter estimates across the three models, and Figure 4 shows the corresponding standardized posterior distributions. The key change appears in the emotion-to-performance pathway, $\beta^{E\to P}$. In both the Hypothesized and Alt1 models, this parameter was credibly negative (Hypothesized: -0.102, 95% CrI $[-0.171, -0.037]$; Alt1: -0.104, 95% CrI $[-0.174, -0.036]$), replicating the result reported by Lin et al. (2026). In the LogTrial model, however, the estimate was substantially attenuated and was no longer credibly different from zero (-0.043, 95% CrI $[-0.114, 0.028]$).

At the same time, a strong positive log(trial) effect on performance emerged in the

Figure 3

Per-Block Pairwise Differences in Expected Log Predictive Density From Exact Leave-Future-Out Cross-Validation (Blocks 6–20). Each bar represents the ΔELPD at a single one-step-ahead prediction step. Blue bars indicate that the first-named model predicted better at that block; red bars indicate the reverse. Annotation text reports the total ΔELPD summed across all 15 blocks and its ratio to the standard error.

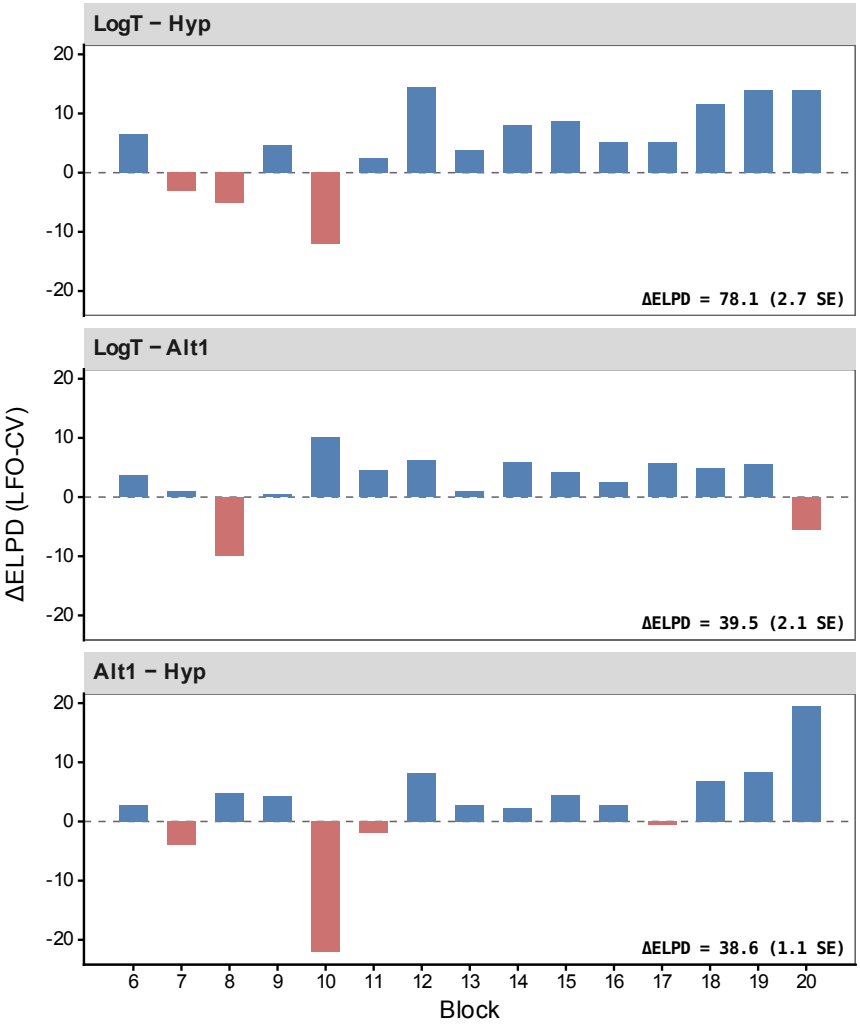


Table 2

Posterior means with 95% credible intervals for population-level parameters across the three models. The Hypothesized model assumes a symmetric GPD effect on goals, whereas Alt1 and LogT estimate separate asymmetric effects conditional on success and failure.

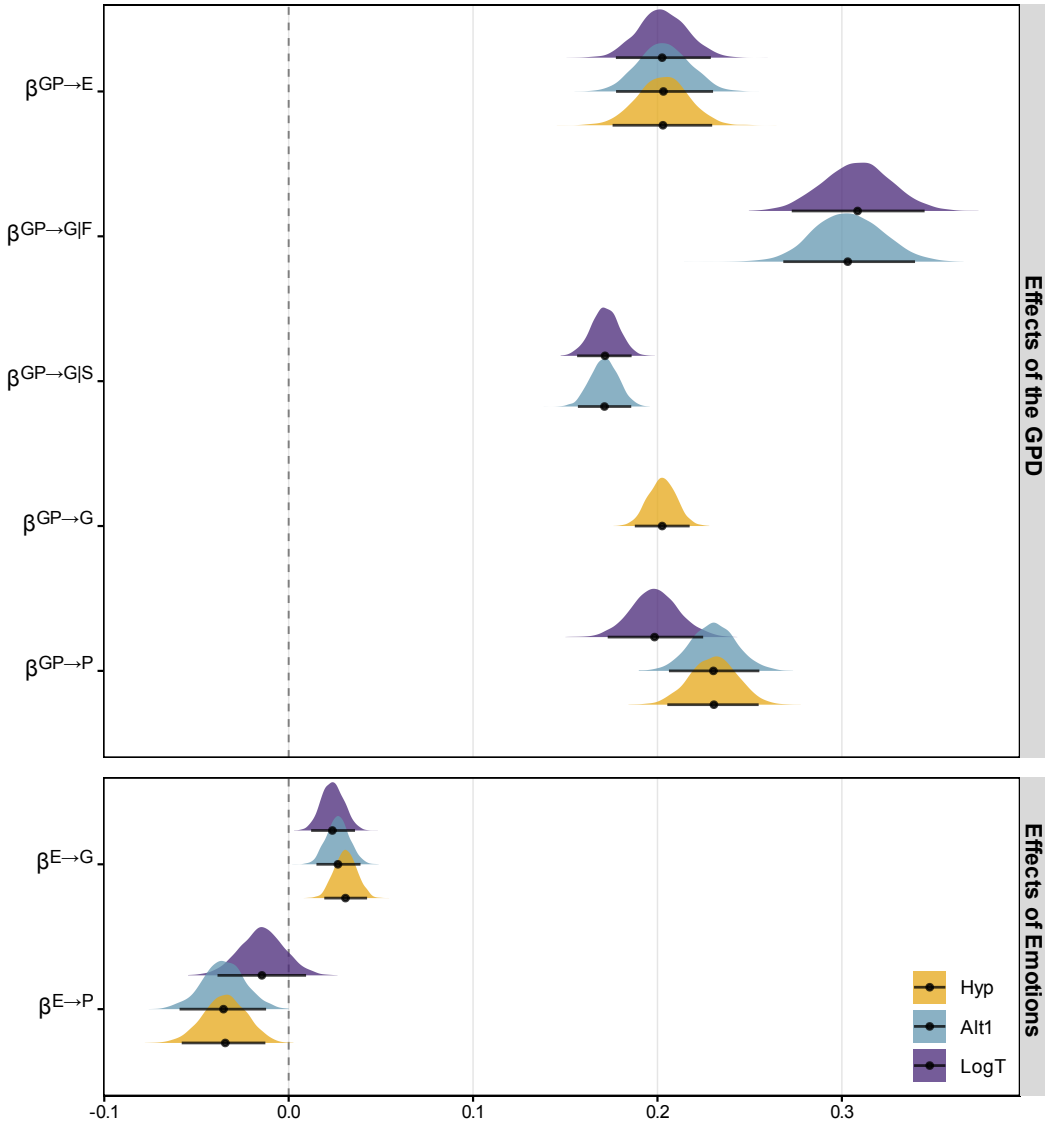
	Estimate [95% CrI]		
Parameter	Model Hyp	Model Alt1	Model LogT
Autoregression effects			
α^G	0.858 [0.838, 0.877]	0.869 [0.850, 0.888]	0.877 [0.857, 0.896]
α^P	0.596 [0.554, 0.637]	0.596 [0.555, 0.636]	0.500 [0.456, 0.544]
α^E	0.515 [0.481, 0.548]	0.515 [0.479, 0.549]	0.513 [0.478, 0.548]
GPD effects			
$\beta^{GP\to G}$	0.322 [0.299, 0.346]	-	-
$\beta^{GP\to G\|S}$	-	0.273 [0.250, 0.296]	0.273 [0.249, 0.296]
$\beta^{GP\to G\|F}$	-	0.483 [0.427, 0.541]	0.491 [0.434, 0.549]
$\beta^{GP\to P}$	0.428 [0.382, 0.474]	0.428 [0.383, 0.475]	0.369 [0.322, 0.418]
$\beta^{GP\to E}$	0.128 [0.111, 0.145]	0.129 [0.112, 0.145]	0.128 [0.112, 0.145]
Emotion effects			
$\beta^{E\to G}$	0.078 [0.049, 0.107]	0.067 [0.038, 0.098]	0.060 [0.031, 0.091]
$\beta^{E\to P}$	-0.102 [-0.171, -0.037]	-0.104 [-0.174, -0.036]	-0.043 [-0.114, 0.028]
log(t) effects			
$\beta_G^{\log t}$	-	-	-0.047 [-0.083, -0.011]
$\beta_P^{\log t}$	-	-	0.326 [0.262, 0.393]
$\beta_E^{\log t}$	-	-	-0.017 [-0.037, 0.002]

LogTrial model ($\beta_P^{\log t} = 0.326$, 95% CrI $[0.262, 0.393]$). Holding the other predictors constant, this implies that participants were expected to improve by roughly one unit of performance from the beginning to the end of the task. If performance is interpreted on the original count scale, this corresponds approximately to one additional correct response. A smaller negative log(trial) effect on goals was also credible ($\beta_G^{\log t} = -0.047$, 95% CrI $[-0.083, -0.011]$). By contrast, the log(trial) effect on emotions was extremely small ($\beta_E^{\log t} = -0.017$, 95% CrI $[-0.037, 0.002]$), suggesting at most a negligible downward trend over time.

Including the learning-curve term also reduced two other coefficients in the performance equation: the autoregressive effect on performance, α^P, decreased from 0.596 in Alt1 to 0.500 in LogTrial, and the GPD effect on performance, $\beta^{GP\to P}$, decreased from 0.428 to 0.369. By contrast, most other parameters remained stable across models. The success-failure asymmetry in goal revision ($\beta^{GP\to G|S} = 0.273$; $\beta^{GP\to G|F} = 0.491$) changed very little from Alt1 to LogTrial, as did the GPD effect on emotions ($\beta^{GP\to E} \approx 0.128$) and the emotion effect on goals, $\beta^{E\to G}$. The

Figure 4

Standardized Population-Level Posterior Distributions of Key Cross-Lagged Parameters Across Three Models. Point estimates and 95% credible intervals are shown. Parameters are standardized by multiplying the raw coefficient by the ratio of predictor to outcome standard deviations. The upper panel displays effects of the goal-performance discrepancy (GPD); $\beta^{GP\rightarrow G}$ appears only for the Hypothesized model (symmetric), while $\beta^{GP\rightarrow G|S}$ (success) and $\beta^{GP\rightarrow G|F}$ (failure) appear for Alt1 and LogTrial. The lower panel displays effects of emotions. The key comparison is $\beta^{E\rightarrow P}$: credible in the Hypothesized and Alt1 models, but no longer credible in the LogTrial model after controlling for practice effects.



main change introduced by log(trial), therefore, was concentrated in the performance equation: once practice was modeled directly, the previously negative emotion-to-performance association was markedly weakened, while a substantial learning effect was estimated instead.

Discussion

The Emotion-to-Performance Pathway: An Omitted Learning Curve in the Performance Equation

The main implication of this reanalysis is that the negative emotion-to-performance effect reported by Lin et al. (2026) was not robust to a minimal and substantively plausible learning-curve specification. In the LogTrial model, $\beta^{E\to P}$ was sharply attenuated and no longer credibly different from zero, while a strong positive learning curve in performance emerged ($\beta_P^{\log t} = 0.326$). Holding the other predictors constant, this learning-curve term implies that participants were expected to improve by roughly one unit of performance from the beginning to the end of the task, corresponding to approximately one additional correct response on the original count scale. Once this practice-related improvement was modeled directly, the empirical support for a negative effect of emotions on subsequent performance largely disappeared.

This substantially weakens the original coasting interpretation. We do not argue that the coasting hypothesis is theoretically wrong. Experimental evidence suggests that positive emotions can, under some conditions, reduce effort (Seo & Patall, 2021; Thürmer et al., 2020). Our claim is narrower and empirical: Study 1 does not provide robust evidence for coasting once a practice-related learning curve is included in the performance equation. The negative emotion-to-performance pathway is better understood as a specification-dependent result than as strong evidence that positive emotions harm later performance.

The pattern of parameter change also helps localize where the confound operated. The coefficients that changed materially were all in the performance equation: the emotion effect weakened, the autoregressive coefficient decreased, and the GPD effect on performance decreased. By contrast, the time trend in emotions was negligible, and the broader self-regulation parameters changed very little. This is exactly the pattern one would expect if the original

performance equation were absorbing variance from a systematic across-trial improvement that had not been modeled explicitly. In other words, the core self-regulation dynamics proposed by Lin et al. (2026), including the role of the GPD in goal revision and emotional response, are not challenged by our reanalysis. What changes is specifically the interpretation of the emotion-to-performance pathway in Study 1.

Competing Influences on the Emotion-to-Performance Pathway

If the negative $\beta^{E\to P}$ in Study 1 was an artifact, the sign reversal between Studies 1 and 2 reported by Lin et al. (2026) may not reflect opposing emotional mechanisms at all. Instead, the estimated $\beta^{E\to P}$ in any given study may reflect the net balance of at least two competing influences: a genuine facilitative effect of positive emotions on performance, consistent with the broader achievement emotions literature (Fredrickson, 2001; Pekrun, 2006), and a practice-related confound that pushes the estimate in the negative direction when a learning curve in performance is unmodeled.

The relative strength of these two influences likely varies with design features. In Study 2, several characteristics of the task may have tilted the balance toward the genuine affective process. The exam carried high personal stakes, plausibly strengthening the motivational relevance of emotions for subsequent effort. The learning content varied across sessions, reducing the scope for a smooth, monotonic practice curve. The temporal spacing of observations over days and weeks further limited within-session accumulation of practice effects. Under such conditions, the practice-related confound would be weaker, allowing a genuine positive emotion-to-performance link to be expressed in the estimate.

Conversely, in a low-stakes repetitive task like Study 1, the genuine affective contribution to performance may be small to begin with: when the task carries little personal significance, how one feels may matter less for how one performs. At the same time, the conditions for a strong practice curve are ideal. The combination of a weak genuine effect and a strong confound would produce a negative $\beta^{E\to P}$ that disappears, as we observed, once the confound is controlled.

This framework suggests that the emotion-to-performance pathway may be most

interpretable in designs where practice-related structure in performance is either inherently weak or explicitly modeled. In highly repetitive paradigms, any estimated $\beta^{E \to P}$ should be treated with caution unless a learning-curve term is included as a robustness check. We note that this account is speculative and should be tested by applying explicit time trends to designs that vary systematically in task repetitiveness and personal relevance.[4]

Model Comparison and Temporal Validation

A second contribution of this reanalysis concerns model comparison. The issue is not simply that Lin et al. (2026) relied on one criterion rather than several; if a single criterion is well matched to the data structure, that can be entirely defensible. The problem here is that the chosen criterion, WAIC, was itself diagnostically unstable in these data, and it was applied in a setting with strong temporal ordering.

That matters empirically. The original preference for Alt1 over the Hypothesized model was already modest under WAIC and PSIS-LOO, and under exact LFO-CV it became small relative to its uncertainty. The LogTrial model was preferred to both alternatives under all criteria, although its advantage over Alt1 was notably smaller under LFO-CV than under PSIS-LOO, consistent with the expected reduction in optimistic bias when future information leakage is eliminated. The substantive conclusion thus depends not only on which model is fit, but also on whether the validation criterion respects the temporal structure of the data.

The broader lesson is about criterion-data alignment. In longitudinal autoregressive settings, approximate criteria such as WAIC and PSIS-LOO may be informative, but they should not be treated as decisive when their diagnostics are poor or when the data have temporal structure that the criterion does not fully respect.

Implications for Computational Modeling of SRL

These results have broader implications for computational modeling in self-regulated learning. First, autoregressive structure does not by itself solve the problem of omitted temporal

[4] We could not test this prediction on Study 2 of Lin et al. (2026) because the data were not publicly available at the time of analysis.

structure. All three models in this study were autoregressive, yet the performance equation still benefited substantially from an explicit learning-curve term. Lagged outcomes capture local persistence, but they do not necessarily capture a global trend driven by practice, habituation, fatigue, or task familiarization. In repeated-measures SRL designs, omitting such structure can distort the interpretation of cross-lagged pathways.

Second, the present case highlights a form of temporal confounding that is distinct from the stable between-person confounds emphasized in recent methodological work on longitudinal models (**?**). Here, the problem was not an unobserved stable trait, but a systematic process tied to measurement order. Such confounding can arise even in well-specified hierarchical autoregressive models if the repeated-task structure itself generates a substantial learning curve in one or more of the modeled outcomes.

Third, the present findings argue for a closer connection between substantive theory and validation strategy. When the theoretical question concerns prediction from temporally ordered data, the preferred model-comparison criterion should preserve that ordering. Otherwise, cross-lagged parameters may appear more decisive than they are.

Limitations

Several limitations should be noted. First, the learning curve was modeled as a population-level log(trial) effect rather than allowing it to vary across individuals. This choice was made for reasons of parsimony and computational stability. Preliminary attempts to estimate random log(trial) slopes led to convergence problems, likely because the number of time points per person was limited relative to the added individual-level complexity. The fixed-effect specification was nevertheless sufficient for the present purpose, namely to test whether a simple learning-curve term altered the key substantive conclusion.

Second, exact LFO-CV is computationally demanding. In our application, with 20 blocks and 15 refits, the cost was manageable (approximately 8 hours per model), but the same approach would become burdensome for substantially longer time series. More efficient alternatives exist in principle. Bürkner et al. (2020) proposed a PSIS approximation to LFO that avoids most refits,

but that method was developed for comparatively simple univariate time-series settings and is not feasible for our jointly autoregressive hierarchical model, where each prediction step adds a large block of new observations and the posterior shift is too large for stable importance sampling. Sequential Monte Carlo methods could in principle recover the one-step-ahead predictive densities required for LFO as a byproduct of a single sequential run, but standard particle-based approaches face severe weight degeneracy in parameter spaces of the scale considered here (approximately 4,000+ parameters). Developing scalable approximations to LFO for complex hierarchical autoregressive models therefore remains an important methodological task.

Third, our empirical reanalysis was limited to Study 1. The interpretation offered for Study 2 is theoretically plausible and generates a clear prediction, but it remains to be tested directly once those data become available.

Fourth, the log(trial) term captures a global nonlinear learning curve but does not identify the underlying mechanism. The observed improvement may reflect practice, familiarity with the interface, procedural adaptation, or some combination of these. Our claim is about omitted learning-curve structure in the performance equation, not about a uniquely identified psychological process.

Conclusion

Our reanalysis supports the broader dynamic framework proposed by Lin et al. (2026): goals, performance, and emotions are interrelated over time, and the goal-performance discrepancy remains central to that system. However, the specific claim that positive emotions negatively affect subsequent performance in Study 1 is not robust once a practice-related learning curve is modeled explicitly. The negative emotion-to-performance coefficient depends on how performance change over trials is specified, and it is substantially reduced once the model is given a direct way to represent across-trial improvement. This reinterpretation also offers a parsimonious account of the otherwise puzzling sign reversal between Studies 1 and 2: rather than opposite emotional mechanisms, the two studies may differ in the balance between a genuine facilitative role of emotions and a practice-related confound that depends on design features such

as task repetitiveness, stakes, and temporal spacing.

References

Ben-Eliyahu, A. (2019). Academic emotional learning: A critical component of self-regulated learning in the emotional learning cycle. *Educational Psychologist*, *54*(2), 84–105. https://doi.org/10.1080/00461520.2019.1582345

Berweger, B., Breitwieser, J., Kracke, B., Brod, G., & Dietrich, J. (2025). Daily affective-motivational experiences during exam preparation: Feedback loops, stability, and change over time. *Journal of Educational Psychology*. https://doi.org/10.1037/edu0000990

Bürkner, P.-C., Gabry, J., & Vehtari, A. (2020). Approximate leave-future-out cross-validation for Bayesian time series models. *Journal of Statistical Computation and Simulation*, *90*(14), 2499–2523. https://doi.org/10.1080/00949655.2020.1783262

Carver, C. (2003). Pleasure as a sign you can attend to something else: Placing positive feelings within a general model of affect. *Cognition and Emotion*, *17*(2), 241–261. https://doi.org/10.1080/02699930302294

Efklides, A. (2011). Interactions of metacognition with motivation and affect in self-regulated learning: The MASRL model. *Educational Psychologist*, *46*(1), 6–25. https://doi.org/10.1080/00461520.2011.538645

Efklides, A., & Schwartz, B. L. (2024). Revisiting the metacognitive and affective model of self-regulated learning: Origins, development, and future directions. *Educational Psychology Review*, *36*(2), 61. https://doi.org/10.1007/s10648-024-09896-9

Forsblom, L., Pekrun, R., Loderer, K., & Peixoto, F. (2022). Cognitive appraisals, achievement emotions, and students' math achievement: A longitudinal analysis. *Journal of Educational Psychology*, *114*(2), 346–367. https://doi.org/10.1037/edu0000671

Fredrickson, B. L. (2001). The role of positive emotions in positive psychology: The broaden-and-build theory of positive emotions. *American Psychologist*, *56*(3), 218–226. https://doi.org/10.1037/0003-066X.56.3.218

Gabry, J., Češnovar, R., Johnson, A., & Bronder, S. (2025). *Cmdstanr: R interface to 'CmdStan'* [Manual]. https://mc-stan.org/cmdstanr/

Jerrim, J. (2022). The power of positive emotions? The link between young people's positive and negative affect and performance in high-stakes examinations. *Assessment in Education: Principles, Policy & Practice*, *29*(3), 310–331. https://doi.org/10.1080/0969594X.2022.2054941

Lin, W. M., FitzGibbon, L., Theobald, M., Breitwieser, J., Brod, G., Murayama, K., & Sakaki, M. (2026). The dynamic interplay between goal setting, performance, and emotions in self-regulated learning: A computational modeling approach. *Journal of Educational Psychology*. https://doi.org/10.1037/edu0001022

Martin, E. A., & Kerns, J. G. (2011). The influence of positive mood on different aspects of cognitive control. *Cognition & Emotion*, *25*(2), 265–279. https://doi.org/10.1080/02699931.2010.491652

Newell, A., & Rosenbloom, P. S. (2013). Mechanisms of skill acquisition and the law of practice: Allen newell and paul S. Rosenbloom. In J. R. Anderson (Ed.), *Cognitive Skills and Their Acquisition* (0th ed., pp. 12–66). Psychology Press. https://doi.org/10.4324/9780203728178-6

Paananen, T., Piironen, J., Bürkner, P.-C., & Vehtari, A. (2021). Implicitly adaptive importance sampling. *Statistics and Computing*, *31*(2), 16. https://doi.org/10.1007/s11222-020-09982-2

Panadero, E. (2017). A review of self-regulated learning: Six models and four directions for research. *Frontiers in Psychology*, *8*, 422. https://doi.org/10.3389/fpsyg.2017.00422

Pekrun, R. (2006). The control-value theory of achievement emotions: Assumptions, corollaries, and implications for educational research and practice. *Educational Psychology Review*, *18*(4), 315–341. https://doi.org/10.1007/s10648-006-9029-9

Pekrun, R., Goetz, T., Titz, W., & Perry, R. P. (2002). Academic emotions in students' self-regulated learning and achievement: A program of qualitative and quantitative

research. In P. A. Schutz & S. L. Lanehart (Eds.), *Emotions in Education* (0th ed., pp. 91–106). Routledge. https://doi.org/10.4324/9781410608628-4

Pekrun, R., Lichtenfeld, S., Marsh, H. W., Murayama, K., & Goetz, T. (2017). Achievement emotions and academic performance: Longitudinal models of reciprocal effects. *Child Development*, *88*(5), 1653–1670. https://doi.org/10.1111/cdev.12704

Pintrich, P. R. (2000). The role of goal orientation in self-regulated learning. In M. Boekaerts, P. R. Pintrich, & M. Zeidner (Eds.), *Handbook of Self-Regulation* (pp. 451–502). Elsevier. https://doi.org/10.1016/B978-012109890-2/50043-3

Rankin, C. H., Abrams, T., Barry, R. J., Bhatnagar, S., Clayton, D. F., Colombo, J., Coppola, G., Geyer, M. A., Glanzman, D. L., Marsland, S., McSweeney, F. K., Wilson, D. A., Wu, C.-F., & Thompson, R. F. (2009). Habituation revisited: An updated and revised description of the behavioral characteristics of habituation. *Neurobiology of Learning and Memory*, *92*(2), 135–138. https://doi.org/10.1016/j.nlm.2008.09.012

Scharfen, J., Peters, J. M., & Holling, H. (2018). Retest effects in cognitive ability tests: A meta-analysis. *Intelligence*, *67*, 44–66. https://doi.org/10.1016/j.intell.2018.01.003

Seo, E., & Patall, E. A. (2021). Feeling proud today may lead people to coast tomorrow: Daily intraindividual associations between emotion and effort in academic goal striving. *Emotion*, *21*(4), 892–897. https://doi.org/10.1037/emo0000752

Tanaka, A., & Murayama, K. (2014). Within-person analyses of situational interest and boredom: Interactions between task-specific perceptions and achievement goals. *Journal of Educational Psychology*, *106*(4), 1122–1134. https://doi.org/10.1037/a0036659

Tenison, C., Fincham, J. M., & Anderson, J. R. (2016). Phases of learning: How skill acquisition impacts cognitive processing. *Cognitive Psychology*, *87*, 1–28. https://doi.org/10.1016/j.cogpsych.2016.03.001

Thompson, R. F., & Spencer, W. A. (1966). Habituation: A model phenomenon for the study of neuronal substrates of behavior. *Psychological Review*, *73*(1), 16–43. https://doi.org/10.1037/h0022681

Thürmer, J. L., Scheier, M. F., & Carver, C. S. (2020). On the mechanics of goal striving: Experimental evidence of coasting and shifting. *Motivation Science*, *6*(3), 266–274. https://doi.org/10.1037/mot0000157

Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, *27*(5), 1413–1432. https://doi.org/10.1007/s11222-016-9696-4

Vehtari, A., Simpson, D., Gelman, A., Yao, Y., & Gabry, J. (2015). *Pareto smoothed importance sampling*. https://doi.org/10.48550/ARXIV.1507.02646

Watanabe, S. (2010). *Asymptotic equivalence of bayes cross validation and widely applicable information criterion in singular learning theory*. https://doi.org/10.48550/ARXIV.1004.2316

Yao, Y., Vehtari, A., Simpson, D., & Gelman, A. (2018). Using stacking to average bayesian predictive distributions (with discussion). *Bayesian Analysis*, *13*(3). https://doi.org/10.1214/17-BA1091

Zimmerman, B. J. (2000). Attaining self-regulation. In *Handbook of Self-Regulation* (pp. 13–39). Elsevier. https://doi.org/10.1016/B978-012109890-2/50031-7

Appendix

Model Specifications

This appendix provides the full mathematical specification of the three models compared in this study. All models share the same hierarchical Bayesian structure and differ only in the specification of the conditional means for goals, performance, and emotions. We first describe the common elements, then present the model-specific equations.

Common Structure

At each time point t (blocks 2 through 20), goals (G), performance (P), and emotions (E) are modeled as jointly normally distributed, conditional on the previous time point:

$$G_{t+1} \sim \mathcal{N}(\mu^G_{t+1},\ \sigma_G), \quad P_{t+1} \sim \mathcal{N}(\mu^P_{t+1},\ \sigma_P), \quad E_{t+1} \sim \mathcal{N}(\mu^E_{t+1},\ \sigma_E) \tag{A1}$$

where σ_G, σ_P, and σ_E are population-level residual standard deviations. Block 1 values serve as initial conditions and are not modeled. Each parameter θ entering the conditional means below is specified hierarchically via a non-centered parameterization: $\theta_i = \bar{\theta} + \sigma_\theta \cdot z_i$, where $z_i \sim \mathcal{N}(0, 1)$, $\bar{\theta} \sim \mathcal{N}(0, 3)$, and $\sigma_\theta \sim \mathcal{N}^+(0, 3)$. Subscript i indexes participants.

Conditional Means

The performance and emotion equations are shared across the Hypothesized and Alt1 models:

$$\mu^P_{t+1} = \gamma^P_i + \alpha^P_i\, P_t + \beta^{GP\to P}_i\, (G_{t+1} - P_t) + \beta^{E\to P}_i\, E_t \tag{A2}$$

$$\mu^E_{t+1} = \gamma^E_i + \alpha^E_i\, E_t + \beta^{GP\to E}_i\, (P_{t+1} - G_{t+1}) \tag{A3}$$

The three models differ in the goal equation and in whether a log-trial trend is included, as described below.

Hypothesized Model. The goal equation uses a single symmetric coefficient for the

goal-performance discrepancy (GPD):

$$\mu_{t+1}^{G} = \gamma_i^G + \alpha_i^G\, G_t + \beta_i^{GP\to G}\,(P_t - G_t) + \beta_i^{E\to G}\, E_t \tag{A4}$$

Alt1 Model. The goal equation replaces the symmetric GPD term with separate coefficients for success and failure:

$$\mu_{t+1}^{G} = \gamma_i^G + \alpha_i^G\, G_t + \beta_i^{E\to G}\, E_t + \begin{cases} \beta_i^{GP\to G|S}\,(P_t - G_t) & \text{if } P_t > G_t \text{ (success)} \\ \beta_i^{GP\to G|F}\,(P_t - G_t) & \text{if } P_t \leq G_t \text{ (failure)} \end{cases} \tag{A5}$$

LogTrial Model. This model extends Alt1 by adding population-level log-trial effects to all three conditional means. The goal equation is identical to Equation A5 with an additional term $\beta_G^{\log t}\log(t+1)$. The performance and emotion equations become:

$$\mu_{t+1}^{P} = \gamma_i^P + \alpha_i^P\, P_t + \beta_i^{GP\to P}\,(G_{t+1} - P_t) + \beta_i^{E\to P}\, E_t + \beta_P^{\log t}\log(t+1) \tag{A6}$$

$$\mu_{t+1}^{E} = \gamma_i^E + \alpha_i^E\, E_t + \beta_i^{GP\to E}\,(P_{t+1} - G_{t+1}) + \beta_E^{\log t}\log(t+1) \tag{A7}$$

The three log-trial coefficients ($\beta_G^{\log t}$, $\beta_P^{\log t}$, $\beta_E^{\log t}$) are population-level only, with no individual-level variation. This choice was made to maintain parsimony; see Limitations for discussion.